Certified Feature Selection for Wasserstein Robust Hypothesis Testing

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Abstract

We study joint feature selection and hypothesis-test design under class-conditional Wasserstein uncertainty. For finite, nonnegative, nondecreasing convex losses, a whole-space test construction yields an exact finite optimization problem. Risk-dependent coordinate prices give a dual representation by transport matrices and correction vectors, producing global lower bounds for every feature subset. The search retains both feasible scores and full dual certificates: old scores give inexpensive feasible upper bounds at new subsets, while old fractions generate additional cuts without a continuous solve. A deterministic acquisition rule favors possible improvement and unresolved risk; a global master supplies certification and a fallback candidate. We prove a quantitative bound on accurate binary evaluations in a data-weighted distance and give an explicit precision-dependent bound from the coordinate movement weights. The argument requires neither differentiability of the value function nor a probabilistic model for it. The certificates concern optimization within the specified uncertainty sets; population error control requires their separate calibration.

1 The selection problem

We can observe at most d of D features. The choice of features and the test using them must be optimized together: a good subset is one on which a test can keep both classwise losses small when the distributions move within their specified uncertainty sets.

Pool the observations from the two classes into $x_1, \dots, x_N \in \mathbb{R}^D$. Observation i has class $c_i \in \{0,1\}$ and weight $q_i > 0$. We encode its class by the sign $y_i = 2c_i - 1$: $y_i = -1$ for class zero and $y_i = +1$ for class one. Both c_i and y_i are fixed, observed labels, not optimization variables. Write $I_k = \{i : c_i = k\}$. The empirical distributions and uncertainty sets are

$$\widehat{P}_k = \sum_{i \in I_k} q_i \delta_{x_i}, \qquad \sum_{i \in I_k} q_i = 1, \qquad \mathcal{U}_k = \{P : W_1(P, \widehat{P}_k) \le \theta_k\}, \quad k = 0, 1.$$

Here δ_x denotes unit mass at x. The distance W_1 is the minimum expected L^1 distance needed to transport one distribution to the other; the distributions have finite first moments. The radii $\theta_k > 0$ and feature scales are fixed before selection. Each class has total weight one, so the pooled weights sum to two.

For an integer budget $0 \le d \le D$, a binary vector z selects $S(z) = \{j : z_j = 1\}$, with $\mathcal{Z}_d = \{z \in \{0,1\}^D : \mathbf{1}^\top z \le d\}$. A measurable score g depends only on the selected coordinates. A positive score favors class zero; the losses are therefore $\Phi(-g)$ for class zero and $\Phi(g)$ for class one. The problem is

$$\begin{aligned} V(z) &= \inf_{g = g(x_{S(z)})} \left\{ \sup_{P \in \mathcal{U}_0} \mathbb{E}_P \Phi(-g) + \sup_{P \in \mathcal{U}_1} \mathbb{E}_P \Phi(g) \right\}, \\ \mathrm{OPT} &= \min_{z \in \mathcal{Z}_d} V(z). \end{aligned} \tag{1}$$

Thus $V(z)$ already includes the best possible test for the chosen features; selection does not hold a previously fitted test fixed.

The function $\Phi : \mathbb{R} \to [0, \infty)$ is finite, convex and nondecreasing. Nondecreasing means that a larger score against the true class cannot reduce its loss. We consider nonconstant losses with $\Phi(0) > 0$. If $\Phi(0) = 0$ or the loss is constant, the constant test already solves the problem. Specific formulas for Φ are not needed for the construction below; examples are given in Section 6. The constant score $g = 0$ provides the common benchmark

$$0 \le V(z) \le B := 2\Phi(0).$$

2 A finite problem that still designs the whole test

The same pooled observations serve as candidate target locations. To compare source i with target ℓ, use one vector of coordinatewise distances:

$$\mathbf{d}_{i\ell} = |x_i - x_\ell| \in \mathbb{R}^D_+, \qquad d_z(x_i, x_\ell) = \mathbf{d}_{i\ell}^\top z.$$

Absolute value is taken coordinatewise. The vector stores the geometry once; its inner product with z is the distance using the selected features.

Let u_ℓ be the score assigned to target ℓ. Let a_i be a loss allowance at source i, and let $\lambda_k \geq 0$ be the price per unit of transport for class k. Write $q = (q_i)$, $\theta = (\theta_0, \theta_1)^\top$, $\lambda = (\lambda_0, \lambda_1)^\top$, and $J(a, \lambda) = q^\top a + \theta^\top \lambda$.

Theorem 1 (Exact finite formulation)**.** *For every binary z,*

$$\begin{aligned} V(z) = \min_{a,u,\lambda} \quad & J(a,\lambda) \\ \textit{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z, \qquad i, \ell = 1, \dots, N, \\ & \lambda \geq 0, \qquad a, u \in \mathbb{R}^N. \end{aligned} \tag{2}$$

The minimum is attained. Every feasible solution yields a measurable test on the entire observation space, using only the selected coordinates, with robust risk at most $J(a, \lambda)$.

The inequality in (2) has a direct meaning: after moving from source i to target ℓ, the loss is at most its source allowance plus the transport price. The source retains its class c_i and sign y_i, while u_ℓ is the score at the target. Explicitly,

$$\Phi(y_i u_\ell) = \begin{cases} \Phi(-u_\ell), & c_i = 0, \\ \Phi(u_\ell), & c_i = 1. \end{cases}$$

Thus the loss uses the source class even when the target observation originally belongs to the other class; either class can send mass to any target location.

The whole-space conclusion is essential. It allows a finite feasible objective to be used as an upper bound for the original test problem. Its proof blends two source scores, one from each class. Convexity controls the blended losses, and the triangle inequality controls the required transport allowances. The construction and both directions of the equality are given in Appendix A; no particular inverse or closed form of Φ is used.

3 Use the risk to control which coordinates are available

Taking $\ell = i$ in (2) gives $a_i \geq \Phi(y_i u_i) \geq 0$. Consequently,

$$\theta^\top \lambda \leq J(a, \lambda). \tag{3}$$

The transport-price contribution is part of the total risk, so it cannot exceed that risk. This inequality supplies the bound needed to connect feature selection with continuous test design.

Introduce two vectors $w_0, w_1 \in \mathbb{R}^D_+$ of coordinate prices. On a binary selection we can take $w_{kj} = \lambda_k z_j$: an unobserved coordinate has price zero. To obtain a useful continuous extension, use the risk bound η to constrain these prices:

$$\begin{aligned} v(z) = \min_{\eta,a,u,\lambda,w_0,w_1} \quad & \eta \\ \text{subject to} \quad & \Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i}, \qquad \forall i, \ell, \\ & J(a,\lambda) \leq \eta, \qquad \lambda \geq 0, \\ & 0 \leq w_k \leq \lambda_k \mathbf{1}, \qquad k = 0, 1, \\ & \theta_0 w_0 + \theta_1 w_1 \leq \eta z. \end{aligned} \tag{4}$$

Vector inequalities hold coordinatewise; a, u, η are real variables. The first line designs the test using the coordinate prices. The next lines bound its risk and keep each coordinate price below the class price. The last line makes an unselected coordinate unavailable and ties the allowed prices to the risk being minimized.

For binary z, an optimizer of (2) enters (4) with $\eta = J$ and $w_k = \lambda_k z$, by (3). Conversely, a feasible point of (4) has zero prices on unselected coordinates and $w_k \le \lambda_k z$. The same a, u, λ therefore satisfy (2), with objective at most η. Hence $v(z) = V(z)$ at every binary selection. For fractional $z \in [0,1]^D$, (4) defines the extension used to construct lower bounds.

Fixing z makes (4) a convex problem. This alone does not prove that its value $v(z)$ is convex in z, since the last line contains the product ηz. The next step establishes that property and supplies information that remains useful when the feature subset changes.

4 One solved subset gives lower bounds for all subsets

Consider a nonnegative matrix Π with $\Pi_{i\ell}$ units of mass sent from source i to target ℓ. Its rows retain the source weights, $\Pi\mathbf{1} = q$. The masses received from the two classes and their coordinatewise movements are

$$p_{k\ell} = \sum_{i \in I_k} \Pi_{i\ell}, \qquad A_k(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} \mathbf{d}_{i\ell} \in \mathbb{R}^D_+.$$

At a target receiving both classes, one score must serve both. The least weighted loss that this common score can achieve is

$$\Psi(p_0, p_1) = \inf_s \{p_0 \Phi(-s) + p_1 \Phi(s)\}, \qquad H(\Pi) = \sum_\ell \Psi(p_{0\ell}, p_{1\ell}). \tag{5}$$

The infimum permits limiting scores when one class has zero mass. H measures the loss left after the best score is chosen at every target. It is nonnegative. Scaling all masses by $t \ge 0$ scales H by t; this property concerns the masses and does not require Φ itself to be homogeneous.

To use a matrix that may move more mass than the original budget allows, introduce a nonnegative correction vector $\mu \in \mathbb{R}^D_+$. The pair (Π, μ) must satisfy

$$\Pi \ge 0, \qquad \Pi\mathbf{1} = q, \qquad \mu \ge 0, \qquad \mathbf{1}^\top \big(A_k(\Pi) - \theta_k \mu\big)_+ \le \theta_k, \quad k = 0, 1. \tag{6}$$

Here $t_+ = \max(t, 0)$, applied coordinatewise to a vector. After $\theta_k \mu_j$ is deducted from movement in coordinate j, the remaining positive movements must fit inside the class budget θ_k. These conditions contain no z: the same pair can be used at any selection. The matrix need not be a transport plan inside the original uncertainty balls. The correction is what makes it useful as a risk lower bound.

The reason for the correction in the denominator can be seen directly. For $0 \le z \le 1$, (6) implies $z^\top A_k(\Pi) \le \theta_k(1 + \mu^\top z)$. For any feasible point of (2) on a binary selection,

$$\begin{aligned} H(\Pi) &\le \sum_{i,\ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ &\le q^\top a + \sum_k \lambda_k z^\top A_k(\Pi) \\ &\le J(a, \lambda) + (\mu^\top z) \theta^\top \lambda \\ &\le (1 + \mu^\top z) J(a, \lambda). \end{aligned}$$

The four steps use, respectively, best-score loss, finite loss constraints, the corrected transport budgets, and (3). Thus $H(\Pi)/(1 + \mu^\top z)$ is a lower bound on $V(z)$.

Theorem 2 (Exact representation and convexity)**.** *For every* $z \in [0,1]^D$,

$$v(z) = \max_{\Pi,\mu \text{ satisfying (6)}} \frac{H(\Pi)}{1+\mu^\top z}. \tag{7}$$

The maximum is attained, v *is convex, and* $v(z) = V(z)$ *on binary selections.*

The preceding inequalities explain why each fraction is a valid lower bound. Equality with v requires more: minimizing the Lagrangian of (4) gives the row masses, transport inequalities and one normalization equation. Rescaling those multipliers gives exactly (7). Appendix B derives the objective and every constraint by eliminating one group of variables at a time. For fixed (Π, μ), the fraction is convex because its numerator is nonnegative and its denominator is a positive affine function. The maximum of these convex lower functions is therefore convex.

5 History-guided search with global certificates

A continuous evaluation provides two reusable objects. Its dual certificate bounds the risk at every selection, while its feasible score supplies a test that can be repriced when the selected coordinates change. We use both objects to decide where another full evaluation is informative.

For a retained feasible pair (Π^s, μ^s), write $h_s = H(\Pi^s)$ and $f_s(z) = h_s/(1+(\mu^s)^\top z)$. At any chosen center $\bar{z} \in [0,1]^D$, its tangent is

$$L_{s,\bar{z}}(z) = \frac{h_s}{1+(\mu^s)^\top \bar{z}} - \frac{h_s(\mu^s)^\top (z-\bar{z})}{[1+(\mu^s)^\top \bar{z}]^2} \le v(z). \tag{8}$$

An optimal certificate touches v at its evaluation point. A feasible certificate still gives a valid lower bound. Retaining the full fraction lets us form additional tangents at promising proposals without solving another continuous problem. Let $\mathcal{C}_t$ contain all these retained tangents and set $m_t(z) = \max\{0, L(z) : L \in \mathcal{C}_t\}$. The global master is the MILP

$$\min_{z \in Z_a, \eta} \eta \quad \text{subject to} \quad \eta \ge 0, \qquad \eta \ge L(z) \quad (L \in \mathcal{C}_t). \tag{9}$$

The lower model used to select candidates is exactly the one used by this master; full fractions strengthen it through new tangents.

Suppose evaluation s also returned a feasible finite score u^s and prices λ^s, with original objective $b_s = J(a^s, \lambda^s) \le B$. At a new binary selection we keep these quantities and recompute only the source allowances:

$$a_i^s(z) = \max_\ell \{\Phi(y_i u_\ell^s) - \lambda_{c_i}^s \mathbf{d}_{i\ell}^\top z\}, \qquad U^s(z) = q^\top a^s(z) + \theta^\top \lambda^s.$$

These allowances satisfy every finite loss constraint. The whole-space construction therefore turns $U^s(z)$ into an original feasible-test bound, including when z differs from the subset on which the score was fitted. Let

$$U_t(z) = \min\{B, U^s(z) : s \text{ is a stored binary evaluation}\}, \qquad W_t(z) = U_t(z) - m_t(z).$$

On binary selections, $m_t(z) \le V(z) \le U_t(z)$. The minimizing stored score supplies the corresponding feasible test; it can improve the incumbent without a full oracle call. A full call at a selected candidate nevertheless refits the entire test and computes a new dual certificate.

The proposal pool may contain coordinate swaps of good subsets, earlier master candidates, or proposals from a separate statistical surrogate. The deterministic acquisition score

$$A_t(z) = (\mathrm{UB}_t - m_t(z))_+ \, W_t(z)$$

favors a subset that remains capable of improving the incumbent and has a wide certified interval. It does not interpret this interval as a predictive variance or require a probability model for V.

To prevent the proposal rule from spending evaluations on almost resolved subsets, fix $0 < \alpha < 1$ and use the global gap $G_t = \mathrm{UB}_t - \mathrm{LB}_t$ as an information threshold. A full evaluation is permitted only when

$$m_t(z) < \mathrm{UB}_t, \qquad W_t(z) > \alpha G_t.$$

A scheduled certification step, or the absence of an eligible proposal, triggers the global master. It supplies a guard candidate z_M and lower bound ℓ_t satisfying $m_t(z_M) - \ell_t \le \xi$. Update the running lower bound with ℓ_t and first update the incumbent using $U_t(z_M)$. Then $W_t(z_M) \ge G_t - \xi$. Between certification steps, the previous running lower bound remains valid. Consequently, $\xi \le (1-\alpha)\varepsilon$ makes this candidate eligible whenever $G_t > \varepsilon$. The pool can choose a more informative candidate, but cannot prevent global exploration. Recentered cuts are added before a guard solve. Its candidate-accuracy condition is used only for that lower model; previous global lower bounds remain valid between guard solves.

Algorithm 1 History-guided certified feature selection

Require: Tolerance $\varepsilon > 0$, $0 < \alpha < 1$; oracle tolerance $0 \le \delta < \alpha\varepsilon$; master tolerance $0 \le \xi \le (1-\alpha)\varepsilon$.

Set $\mathrm{LB} = 0$, $\mathrm{UB} = B$; store the constant feasible test, and initialize the cut and evaluation histories.
while $\mathrm{UB} - \mathrm{LB} > \varepsilon$ **do**
    Build a proposal pool from the history and optionally recenter stored fractions there; retain every resulting valid tangent.
    Reprice stored scores at the pool's binary proposals and update UB with the best resulting feasible test; set $G = \mathrm{UB} - \mathrm{LB}$ and stop if $G \le \varepsilon$.
    Form the eligible proposals with $m(z) < \mathrm{UB}$ and $W(z) > \alpha G$.
    if certification is scheduled or no proposal is eligible **then**
        Solve (9) to obtain a guard z_M and certified lower bound ℓ with $m(z_M) - \ell \le \xi$; set $\mathrm{LB} \leftarrow \max\{\mathrm{LB}, \ell\}$.
        Add z_M to the pool, reprice it and update UB. Recompute G and eligibility; stop if $G \le \varepsilon$.
    end if
    Choose an eligible proposal maximizing $A(z)$; a refreshed guard guarantees eligibility when it is needed.
    Fully evaluate (4) at this binary z. Retain the best genuine feasible test and a retained valid tangent L_s such that their anchor gap satisfies $b_s - L_s(z) \le \delta$.
    Store that test's original finite score and prices, the available dual fractions and L_s; update UB with the best feasible test.
    If a time or accuracy budget is exhausted, return the available bounds and test without claiming the convergence conclusions below.
end while
return The best feasible selection and test, with [LB, UB].

An evaluation here means obtaining the stated anchor accuracy δ; it may require several numerical refinements. The count below concerns completed accuracy-controlled binary evaluations, not fractional root calls, individual LP iterations, or wall time.

The quantitative guarantee uses the geometry already present in the data. Define

$$M_j = \max_{k=0,1} \frac{\sum_{i \in I_k} q_i \max_\ell d_{i\ell,j}}{\theta_k}, \qquad \rho(z, z') = \sum_j M_j |z_j - z'_j|.$$

A coordinate with small M_j can cause only a small movement relative to the uncertainty budgets. The function ρ is a weighted Hamming distance on binary vectors; zero weights are allowed. In (7), the

correction can be restricted to $0 \le \mu \le M$: clipping at M preserves (6) and raises the fraction. This restriction is applied to the pair (Π, μ) whose row masses equal q, before generating its tangents. For a finite set $\mathcal{Z}$, let $\mathcal{P}(\mathcal{Z}, \rho, r)$ be the largest number of its points whose pairwise distances are strictly greater than r. To express the evaluation bound directly in data quantities, sort $M_{(1)} \ge \cdots \ge M_{(D)}$ and define

$$r_\varepsilon = \frac{\alpha\varepsilon - \delta}{B}, \qquad s_\varepsilon = \min\left\{ s \in \{0, \dots, D\} : \sum_{j>s} M_{(j)} \le r_\varepsilon \right\}.$$

Theorem 3 (Global bounds and quantitative convergence)**.** *Every retained tangent and every repriced test are valid for the original problem, and the running bounds satisfy*

$$\mathrm{LB}_t \le \mathrm{OPT} \le \mathrm{UB}_t. \tag{10}$$

Suppose Algorithm 1 completes its required solves without a premature time or accuracy exit. After $N \ge 1$ *full binary evaluations, the history satisfies*

$$W_N(z) \le \delta + B \min_{1 \le s \le N} \rho(z, z^s).$$

At a refreshed guard, its certified global gap satisfies

$$G_N \le \delta + \xi + B \min_{1 \le s \le N} \rho(z_M, z^s).$$

If $G_N > \delta/\alpha$*, the* N *evaluated selections are pairwise separated by more than* $(\alpha G_N - \delta)/B$*, and hence*

$$N \le \mathcal{P}\left(\mathcal{Z}_d, \rho, \frac{\alpha G_N - \delta}{B}\right).$$

In particular, the number N_ε *of completed binary evaluations begun while the gap exceeds* ε *satisfies*

$$N_\varepsilon \le \mathcal{P}(\mathcal{Z}_d, \rho, r_\varepsilon) \le \min\left\{ |\mathcal{Z}_d|, \sum_{i=0}^{\min(d, s_\varepsilon)} \binom{s_\varepsilon}{i} \right\} \le 2^{s_\varepsilon}.$$

The algorithm therefore reaches an ε*-optimal feasible test.*

Appendix C proves these statements from one-sided changes in the retained bounds. The factor is B, rather than $2B$: removing coordinates can increase the transferred upper bound, while adding coordinates can decrease an anchor tangent, and the two contributions occupy opposite coordinate directions.

The computable quantity s_ε is the number of largest movement weights needed to control the omitted-coordinate tail at the requested precision. It is not a ranking of statistical feature importance. An r_ε-separated collection must have distinct patterns on these coordinates, which gives the explicit evaluation bound. At very fine precision, s_ε can approach D and the bound remains combinatorial. The appendix also states a conditional rate when an additional packing estimate is available; no such condition is assumed for arbitrary datasets. Adding coordinates cannot worsen V; when features have no extra charge, the implementation may restrict its binary search to $\mathbf{1}^\top z = d$.

6 Loss choices and the scope of the result

The loss enters through Φ in the continuous problem and through Ψ in the dual objective. Examples covered by the same derivation are shown below; boundary values use continuous limits.

Loss	$\Phi(s)$	$\Psi(p_0, p_1)$
Half hinge	$\frac{1}{2}(1+s)_+$	$\min(p_0, p_1)$
Logistic	$\log(1+e^s)$	$(p_0+p_1)\log(p_0+p_1) - p_0\log p_0 - p_1\log p_1$
Exponential	e^s	$2\sqrt{p_0 p_1}$
Squared hinge	$(1+s)_+^2$	$4p_0p_1/(p_0+p_1)$

For example, half-hinge masses $p_0 = 0.2$ and $p_1 = 0.5$ leave loss 0.2: the best common score favors the larger mass but cannot distinguish the smaller mass at the same location. Appendix D derives all four formulas. For half hinge, a score clipped to $[-1, 1]$ gives randomized class-zero probability $(1 + g)/2$, so $V/2$ is the equal-prior worst-case error. Other losses define surrogate objectives. The current implementation uses half hinge and logistic; the other rows illustrate the theoretical scope.

The loss formulas need not be fixed before deriving the algorithm. The transport geometry still matters: the whole-space construction uses the triangle inequality, and the coordinate-price formulation uses the L^1 decomposition. The certified interval concerns the stated robust optimization problem; whether its uncertainty sets cover an unknown population is a separate statistical question.

A Proof of Theorem 1: the finite formulation

Use the assumptions of Section 1, and let $T(z)$ denote the infimum of the finite program on the right of (2). We prove both value inequalities and construct a whole-space test, then establish attainment.

A.1 Finite transports give the lower value bound

Fix binary z and $S = \{j : z_j = 1\}$. Write P_S for the distribution of the selected coordinates under P. The selected distance, projection, and lifting satisfy

$$\begin{aligned} d_z(x,x') &:= \sum_{j=1}^{D} z_j |x_j - x'_j| = \|x_S - x'_S\|_1 \le \|x - x'\|_1, \\ W_1(P_S, \widehat{P}_{k,S}) &\le W_1(P, \widehat{P}_k), \\ \|x_i - (x_{\ell,S}, x_{i,S^c})\|_1 &= d_z(x_i, x_\ell), \\ g(x_{\ell,S}, x_{i,S^c}) &= g(x_\ell). \end{aligned} \tag{A.1}$$

Lifting retains the unselected source coordinates, so it preserves both selected transport cost and test loss. If source projections coincide, split their transported mass according to their weights and retain their source identities. The same construction lifts arbitrary projected targets and preserves finite first moments.

For scores $u = (u_\ell)_{\ell=1}^N$, define $Q_k(u)$ as the largest class-k loss obtained by sending mass to the projected pool. The variable $\Pi_{i\ell}$ sends mass from source $i \in I_k$ to target ℓ; only class-k rows occur in this calculation:

$$\begin{aligned} Q_k(u) = \max_{\Pi \ge 0} \quad & \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) \\ \text{subject to} \quad & \sum_{\ell} \Pi_{i\ell} = q_i \quad (i \in I_k), \\ & \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \le \theta_k. \end{aligned} \tag{A.2}$$

Staying at the source is feasible, and $0 \le \Pi_{i\ell} \le q_i$ makes the feasible set compact. To derive the dual, assign unrestricted row multipliers a_i and a nonnegative budget multiplier λ_k. Because this is a maximization problem, the Lagrangian is

$$\begin{aligned} \mathcal{L}_k &= \sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i u_\ell) + \sum_{i \in I_k} a_i \left(q_i - \sum_{\ell} \Pi_{i\ell} \right) \\ &\quad + \lambda_k \left(\theta_k - \sum_{i \in I_k, \ell} \Pi_{i\ell} d_z(x_i, x_\ell) \right) \\ &= \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k + \sum_{i \in I_k, \ell} \Pi_{i\ell} [\Phi(y_i u_\ell) - a_i - \lambda_k d_z(x_i, x_\ell)]. \end{aligned} \tag{A.3}$$

For a feasible transport the row terms vanish and the budget term is nonnegative, so this Lagrangian bounds its loss above. After these constraints enter the Lagrangian, its supremum retains only $\Pi \ge 0$. A positive mass coefficient gives $+\infty$; if every coefficient is nonpositive, $\Pi = 0$ attains the supremum:

$$\sup_{\Pi \ge 0} \mathcal{L}_k = \begin{cases} \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k, & \Phi(y_i u_\ell) \le a_i + \lambda_k d_z(x_i, x_\ell) \quad (i \in I_k,\ \forall \ell), \\ +\infty, & \text{otherwise.} \end{cases} \tag{A.4}$$

Minimize this upper bound over $a_i \in \mathbb{R}$ and $\lambda_k \geq 0$. Finite linear programming strong duality gives

$$Q_k(u) = \min_{\substack{a_i \in \mathbb{R},\ \lambda_k \geq 0 \\ \Phi(y_i u_\ell) \leq a_i + \lambda_k d_z(x_i, x_\ell)\ (i \in I_k,\ \forall \ell)}} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right]. \tag{A.5}$$

The two classes share u, so the last equality gives $T(z) = \inf_u [Q_0(u) + Q_1(u)]$. For any original test, define $u(g) = (g(x_1), \ldots, g(x_N))$. Lift each feasible transport using (A.1): its full-space distribution lies in $\mathcal{U}_k$ and its loss is $\sum_{i \in I_k, \ell} \Pi_{i\ell} \Phi(y_i g(x_\ell))$. Thus

$$Q_k(u(g)) \leq \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g). \tag{A.6}$$

Restricting the infimum to induced scores and using this inequality gives

$$\begin{aligned} T(z) &= \inf_u [Q_0(u) + Q_1(u)] \\ &\leq \inf_g [Q_0(u(g)) + Q_1(u(g))] \leq V(z). \end{aligned} \tag{A.7}$$

Separate scores at coincident projected targets only relax the first infimum.

A.2 A whole-space test gives the reverse value bound

Take any feasible (a, u, λ) in (2). To satisfy every source allowance at each x, we seek a test with class-k loss below their minimum:

$$\begin{aligned} E_k(x) &= \min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}, \\ i_k(x) &= \min \arg\min_{i \in I_k} \{a_i + \lambda_k d_z(x_i, x)\}. \end{aligned} \tag{A.8}$$

Choose the smallest source index in a tie. At fixed x, abbreviate $i_k = i_k(x)$; c is the distance between the chosen sources and r_k their distances to x:

$$c = d_z(x_{i_0}, x_{i_1}), \qquad r_k = d_z(x_{i_k}, x), \qquad c \leq r_0 + r_1. \tag{A.9}$$

When $c > 0$, blend their scores by

$$\alpha = \frac{\min(c, r_0)}{c}, \qquad g(x) = (1-\alpha) u_{i_0} + \alpha u_{i_1}. \tag{A.10}$$

The triangle inequality ensures that both portions of this distance fit within r_0, r_1:

$$\begin{aligned} 0 \leq \alpha \leq 1, \qquad & \alpha c = \min(c, r_0) \leq r_0, \\ & (1-\alpha)c = c - \min(c, r_0) = \max(c - r_0, 0) \leq r_1. \end{aligned} \tag{A.11}$$

In particular, the last inequality uses $c - r_0 \leq r_1$ and $r_1 \geq 0$.

Substitute source–target pairs (i_0, i_0), (i_0, i_1), (i_1, i_0), and (i_1, i_1) in (2). Their source signs are -1 and $+1$, and the own-source distances are zero:

$$\begin{aligned} \Phi(-u_{i_0}) &\leq a_{i_0}, & \Phi(-u_{i_1}) &\leq a_{i_0} + \lambda_0 c, \\ \Phi(u_{i_0}) &\leq a_{i_1} + \lambda_1 c, & \Phi(u_{i_1}) &\leq a_{i_1}. \end{aligned} \tag{A.12}$$

Convexity, these four constraints, and (A.11) give the two pointwise loss bounds:

$$\begin{aligned} \Phi(-g(x)) &\leq (1-\alpha)\Phi(-u_{i_0}) + \alpha \Phi(-u_{i_1}) \\ &\leq (1-\alpha) a_{i_0} + \alpha(a_{i_0} + \lambda_0 c) = a_{i_0} + \lambda_0 \alpha c \\ &\leq a_{i_0} + \lambda_0 r_0 = E_0(x), \\ \Phi(g(x)) &\leq (1-\alpha)\Phi(u_{i_0}) + \alpha \Phi(u_{i_1}) \\ &\leq (1-\alpha)(a_{i_1} + \lambda_1 c) + \alpha a_{i_1} = a_{i_1} + \lambda_1 (1-\alpha) c \\ &\leq a_{i_1} + \lambda_1 r_1 = E_1(x). \end{aligned} \tag{A.13}$$

If $c = 0$, set $g(x) = u_{i_0}$. The first and third endpoint inequalities give instead

$$\Phi(-g(x)) \le a_{i_0} \le E_0(x), \qquad \Phi(g(x)) \le a_{i_1} + \lambda_1 c = a_{i_1} \le E_1(x). \tag{A.14}$$

Finite comparisons of continuous functions make $i_k(x)$ Borel measurable, and hence the piecewise g is Borel measurable. It uses only x_S and lies between $\min_\ell u_\ell$ and $\max_\ell u_\ell$. It extends the loss bounds; interpolation of every prescribed u_ℓ is unnecessary.

Now integrate these bounds. Fix $P \in \mathcal{U}_k$ and choose a coupling of cost at most θ_k, retaining source indices and splitting coincident atoms according to their weights. Let γ_k be its measure on $I_k \times \mathbb{R}^D$, with target marginal P. Its source rows and cost satisfy

$$\gamma_k(\{i\} \times \mathbb{R}^D) = q_i, \qquad \int d_z(x_i, x)\, d\gamma_k(i,x) \le \int \|x_i - x\|_1\, d\gamma_k(i,x) \le \theta_k. \tag{A.15}$$

For each source $i \in I_k$, the constructed score satisfies $\Phi((2k-1)g(x)) \le E_k(x) \le a_i + \lambda_k d_z(x_i, x)$. Using both marginals gives

$$\begin{aligned}
\mathbb{E}_P \Phi((2k-1)g) &= \int \Phi((2k-1)g(x))\, d\gamma_k(i,x) \\
&\le \int a_i\, d\gamma_k(i,x) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&= \sum_{i \in I_k} a_i \gamma_k(\{i\} \times \mathbb{R}^D) + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&= \sum_{i \in I_k} q_i a_i + \lambda_k \int d_z(x_i, x)\, d\gamma_k(i,x) \\
&\le \sum_{i \in I_k} q_i a_i + \theta_k \lambda_k.
\end{aligned} \tag{A.16}$$

The bound is independent of P, so taking the classwise suprema and adding yields

$$\sum_{k=0}^{1} \sup_{P \in \mathcal{U}_k} \mathbb{E}_P \Phi((2k-1)g) \le \sum_{k=0}^{1} \left[\sum_{i \in I_k} q_i a_i + \theta_k \lambda_k \right] = J(a, \lambda).$$

Every finite feasible point therefore yields an original test with risk at most J. Taking the infimum gives $V(z) \le T(z)$; (A.7) gives the other direction. This proves the value equality and feasible-test claim in Theorem 1.

A.3 Attainment of the primal minima

The following argument proves attainment for Theorem 1 and for the risk program used in Theorem 2. Nonconstancy and monotonicity give $s_1 < s_2$ with $\Phi(s_2) > \Phi(s_1)$. Set $b = [\Phi(s_2) - \Phi(s_1)]/(s_2 - s_1) > 0$; convexity gives, for $t > s_2$,

$$\frac{\Phi(t) - \Phi(s_2)}{t - s_2} \ge b \quad \Longrightarrow \quad \Phi(t) \ge \Phi(s_2) + b(t - s_2) \longrightarrow \infty. \tag{A.17}$$

The constant point $u = 0$, $a_i = \Phi(0)$, $\lambda = 0$ is feasible in (2) with $J = B$. Adding $w_0 = w_1 = 0$ and $\eta = B$ makes it feasible in (4) for every $z \in [0,1]^D$. Thus the objective sublevels $J \le B$ and $J \le \eta \le B$ are nonempty. Own-source constraints give $a_i \ge \Phi(y_i u_i) \ge 0$; positive weights and radii then give

$$\begin{aligned}
0 \le q_i a_i \le q^\top a \le J \le B \quad &\Longrightarrow\ 0 \le a_i \le B/q_i, \\
0 \le \theta_k \lambda_k \le \theta^\top \lambda \le J \le B \quad &\Longrightarrow\ 0 \le \lambda_k \le B/\theta_k, \\
0 \le w_{kj} \le \lambda_k \le B/\theta_k, \quad &0 \le J \le \eta \le B \quad \text{in (4).}
\end{aligned} \tag{A.18}$$

Choose a fixed source $i_k \in I_k$ for each class. The finite pool distances and (A.18) bound both score signs by the finite constant

$$C = \max_{k,\ell} \left\{ B/q_{i_k} + (B/\theta_k)\|\mathbf{d}_{i_k\ell}\|_1 \right\},$$

$$\Phi((2k-1)u_\ell) \le B/q_{i_k} + (B/\theta_k)\|\mathbf{d}_{i_k\ell}\|_1 \le C.$$

Choose $R > 0$ with $\Phi(R) > C$, possible by (A.17). Monotonicity gives

$$u_\ell \ge R \implies \Phi(u_\ell) > C, \qquad u_\ell \le -R \implies \Phi(-u_\ell) > C; \qquad |u_\ell| < R.$$

All variables on these sublevels are bounded. The finite convex loss is continuous, so the sublevels are closed and hence compact. Both continuous objectives attain their minima there. Since a constant test with value B is already feasible, restricting to these sublevels preserves both global minima.

B Proof of Theorem 2: the dual representation

B.1 Multipliers and the Lagrangian

Fix $z \in [0,1]^D$ in (4). The variables η, a, u are free, whereas λ, w_0, w_1 remain nonnegative. Write each remaining constraint with left side at most zero and attach the multiplier shown below:

Constraint	Nonnegative multiplier
$\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} \le 0$	$\bar{\Pi}_{i\ell}$
$J - \eta \le 0$	τ
$w_{kj} - \lambda_k \le 0$	$\bar{\beta}_{kj}$
$\theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j \le 0$	$\bar{\mu}_j$

Here $\bar{\Pi} \in \mathbb{R}_+^{N\times N}$, $\tau \in \mathbb{R}_+$, and $\bar{\beta}_k, \bar{\mu} \in \mathbb{R}_+^D$. The bars distinguish these multipliers from the row-normalized matrix and correction vector in the main text.

Adding multiplier times constraint to the objective gives

$$\begin{aligned} \mathcal{L} = \eta + \tau(J-\eta) &+ \sum_{i,\ell} \bar{\Pi}_{i\ell}[\Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i}] \\ &+ \sum_k \bar{\beta}_k^\top (w_k - \lambda_k \mathbf{1}) + \bar{\mu}^\top(\theta_0 w_0 + \theta_1 w_1 - \eta z). \end{aligned} \tag{B.1}$$

At a feasible point of (4), every added constraint term is nonpositive. Therefore, with the multipliers fixed,

$$\inf_{\substack{\eta\in\mathbb{R},\ a,u\in\mathbb{R}^N \\ \lambda,w_0,w_1\ge 0}} \mathcal{L} \le \mathcal{L}(\eta, a, u, \lambda, w_0, w_1) \le \eta.$$

The infimum is a lower bound on $v(z)$. The dual problem maximizes this lower bound over the multipliers; equality will be verified below. All four constraint groups are now in $\mathcal{L}$; they are not imposed again when taking its infimum. In particular, the nonnegativity of a and η implied by the original constraints does not restrict their domains in this infimum.

To collect the score and price terms, define

$$\bar{p}_{k\ell} = \sum_{i\in I_k} \bar{\Pi}_{i\ell}, \qquad A_k(\bar{\Pi}) = \sum_{i\in I_k,\ell} \bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}. \tag{B.2}$$

The first quantity is the mass of class k at target ℓ; the second is its vector of coordinate movements. The regrouping uses

$$\begin{aligned}\sum_{i,\ell}\bar{\Pi}_{i\ell}a_i &= \sum_i a_i \sum_\ell \bar{\Pi}_{i\ell},\\ \sum_{i,\ell}\bar{\Pi}_{i\ell}\Phi(y_i u_\ell) &= \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)],\\ \sum_{i,\ell}\bar{\Pi}_{i\ell}\mathbf{d}_{i\ell}^\top w_{c_i} &= \sum_k w_k^\top A_k(\bar{\Pi}),\\ \tau J &= \sum_i \tau q_i a_i + \sum_k \tau\theta_k\lambda_k.\end{aligned} \tag{B.3}$$

Thus each primal variable group has its own term:

$$\begin{aligned}\mathcal{L} = \eta(1-\tau-\bar{\mu}^\top z) &+ \sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell})\\ &+ \sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)]\\ &+ \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top\bar{\beta}_k) + \sum_k w_k^\top[\bar{\beta}_k + \theta_k\bar{\mu} - A_k(\bar{\Pi})].\end{aligned} \tag{B.4}$$

B.2 Minimizing over the primal variables

Keep all multipliers fixed. The dual value is the infimum of (B.4) over $\eta \in \mathbb{R}$, $a, u \in \mathbb{R}^N$ and $\lambda, w_0, w_1 \geq 0$. Since these domains are independent, minimize each term separately.

For a linear coefficient b, the two rules are

$$\inf_{t\in\mathbb{R}} bt = \begin{cases}0, & b = 0,\\ -\infty, & b \neq 0,\end{cases} \qquad \inf_{t\geq 0} bt = \begin{cases}0, & b \geq 0,\\ -\infty, & b < 0.\end{cases}$$

For the first rule, $t = -nb$ gives $bt = -nb^2 \to -\infty$ when $b \neq 0$. For the second, $t = n$ gives $bt \to -\infty$ when $b < 0$; otherwise the minimum is zero at $t = 0$.

The risk variable and source allowances. Applying the free-variable rule gives

$$\begin{aligned}\inf_{\eta\in\mathbb{R}} \eta(1-\tau-\bar{\mu}^\top z) > -\infty \iff 1-\tau-\bar{\mu}^\top z = 0,\\ \inf_{a\in\mathbb{R}^N}\sum_i a_i(\tau q_i - \sum_\ell \bar{\Pi}_{i\ell}) > -\infty \iff \tau q_i - \sum_\ell \bar{\Pi}_{i\ell} = 0 \quad (\forall i).\end{aligned} \tag{B.5}$$

Both terms then contribute zero. The second set of equalities is $\bar{\Pi}\mathbf{1} = \tau q$.

The scores. One score u_ℓ serves the two masses at target ℓ. By the definition of Ψ in (5),

$$\begin{aligned}\inf_{u\in\mathbb{R}^N}\sum_\ell [\bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell)] &= \sum_\ell \inf_{s\in\mathbb{R}}[\bar{p}_{0\ell}\Phi(-s) + \bar{p}_{1\ell}\Phi(s)]\\ &= \sum_\ell \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) = H(\bar{\Pi}).\end{aligned} \tag{B.6}$$

To justify the first equality even at boundary masses, fix $\varepsilon > 0$ and choose each u_ℓ so that

$$\Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) \le \bar{p}_{0\ell}\Phi(-u_\ell) + \bar{p}_{1\ell}\Phi(u_\ell) \le \Psi(\bar{p}_{0\ell}, \bar{p}_{1\ell}) + \varepsilon/N.$$

Summing puts the loss between $H(\bar{\Pi})$ and $H(\bar{\Pi}) + \varepsilon$; let $\varepsilon \downarrow 0$. The infima are finite because

$$0 \le \Psi(p_0, p_1) \le \Phi(0)(p_0 + p_1), \qquad 0 \le H(\bar{\Pi}) \le \Phi(0) \sum_{i,\ell} \bar{\Pi}_{i\ell} = \tau B.$$

The last equality uses the row condition and $\sum_i q_i = 2$.

The class and coordinate prices. Applying the nonnegative-variable rule gives

$$\begin{aligned} \inf_{\lambda \ge 0} \sum_k \lambda_k(\tau\theta_k - \mathbf{1}^\top \bar{\beta}_k) > -\infty \iff \mathbf{1}^\top \bar{\beta}_k \le \tau\theta_k \quad (\forall k), \\ \inf_{w_0, w_1 \ge 0} \sum_k w_k^\top [\bar{\beta}_k + \theta_k \bar{\mu} - A_k(\bar{\Pi})] > -\infty \iff A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k \bar{\mu} \quad (\forall k). \end{aligned} \tag{B.7}$$

Each term contributes zero when its condition holds. Therefore the minimized Lagrangian is $H(\bar{\Pi})$ exactly under (B.5) and (B.7), and is $-\infty$ otherwise. Maximizing this finite value over nonnegative multipliers is the dual problem.

It remains to remove $\bar{\beta}_k$. Coordinatewise, its conditions imply

$$\begin{aligned} \bar{\beta}_{kj} \ge 0, \quad A_{kj}(\bar{\Pi}) \le \bar{\beta}_{kj} + \theta_k \bar{\mu}_j \iff \bar{\beta}_{kj} \ge \max(0, A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j), \\ \mathbf{1}^\top \bar{\beta}_k \le \tau\theta_k \implies \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \le \tau\theta_k. \end{aligned} \tag{B.8}$$

Conversely, if the last inequality holds, choose

$$\bar{\beta}_k = (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+.$$

This vector is nonnegative, has sum at most $\tau\theta_k$, and satisfies $A_k(\bar{\Pi}) \le \bar{\beta}_k + \theta_k \bar{\mu}$. Thus both directions are exact, and the dual is

$$\begin{aligned} \max_{\tau, \bar{\Pi}, \bar{\mu} \ge 0} \quad & H(\bar{\Pi}) \\ \text{subject to} \quad & \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z = 1, \\ & \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \le \tau\theta_k, \quad k = 0, 1. \end{aligned} \tag{B.9}$$

We next verify equality with the primal optimum and attainment, before dividing by τ.

B.3 Equality of primal and dual values

Primal attainment was established in Appendix A.3.

The restriction to objective values at most B was used only to prove existence; it is not added to (4). For strong duality in that program, first identify the prices already fixed by a zero coordinate:

$$z_j = 0 \implies 0 \le \theta_0 w_{0j} + \theta_1 w_{1j} \le \eta z_j = 0 \implies w_{0j} = w_{1j} = 0.$$

This uses $\theta_k > 0$ and $w_{kj} \ge 0$. Delete these fixed variables and the resulting tautological coordinate inequalities. On the remaining coordinates choose

$$u = 0, \quad a_i = \Phi(0) + 1, \quad \lambda_k = 1, \quad w_k = \delta z, \quad 0 < \delta < 1, \quad \eta = J + 1 + \delta(\theta_0 + \theta_1). \tag{B.10}$$

Here $J = 2[\Phi(0) + 1] + \theta_0 + \theta_1$. Every relevant residual is strictly negative:

$$\begin{aligned} \Phi(y_i u_\ell) - a_i - \mathbf{d}_{i\ell}^\top w_{c_i} &= -1 - \delta \mathbf{d}_{i\ell}^\top z < 0, \\ w_{kj} - \lambda_k &= \delta z_j - 1 < 0, \\ J - \eta &= -1 - \delta(\theta_0 + \theta_1) < 0, \\ \theta_0 w_{0j} + \theta_1 w_{1j} - \eta z_j &= -(J+1) z_j < 0 \quad (z_j > 0). \end{aligned} \tag{B.11}$$

Also $w_{kj} > 0$ on those coordinates and $\lambda_k > 0$. Finite-dimensional convex strong duality applies to this reduced problem: the dual maximum is attained and equals $v(z)$.

To restore an omitted coordinate, use

$$z_j = 0: \quad \bar{\beta}_{kj} = 0, \qquad \bar{\mu}_j = \max_k A_{kj}(\bar{\Pi})/\theta_k.$$

Then $\bar{\beta}_{kj} + \theta_k \bar{\mu}_j \geq A_{kj}(\bar{\Pi})$ for both classes, no allocation budget increases, and $\bar{\mu}_j z_j = 0$ leaves normalization unchanged. Thus the full dual (B.9) also has an attained maximum equal to $v(z)$.

B.4 Removing the normalization

We first justify division by τ. If $\tau = 0$, nonnegativity and the row condition give

$$\bar{\Pi}\mathbf{1} = 0 \implies \bar{\Pi} = 0 \implies H(\bar{\Pi}) = \sum_\ell \inf_s 0 = 0.$$

For a positive scale, the mass-scaling identity follows directly from the infimum:

$$\begin{aligned} \Psi(\tau p_0, \tau p_1) &= \inf_s \tau[p_0 \Phi(-s) + p_1 \Phi(s)] = \tau \Psi(p_0, p_1), \\ H(\tau \Pi) &= \sum_\ell \Psi(\tau p_{0\ell}, \tau p_{1\ell}) = \tau H(\Pi), \qquad \tau > 0. \end{aligned} \tag{B.12}$$

There is, however, a feasible point with strictly positive objective. Choose a target ℓ_*, and let e_{ℓ_*} be the vector with entry one at that target and zero elsewhere. Sending every source to this target means $\Pi = q e_{\ell_*}^\top$. Its received masses are one from each class, so

$$H(\Pi) = \Psi(1,1), \qquad \Phi(-s) + \Phi(s) \geq 2\Phi(0), \qquad \Psi(1,1) = 2\Phi(0) = B > 0.$$

The last equality follows by taking $s = 0$. Set

$$\mu_j = \max_k \frac{A_{kj}(\Pi)}{\theta_k}, \qquad \tau = \frac{1}{1 + \mu^\top z}, \qquad \bar{\Pi} = \tau \Pi, \qquad \bar{\mu} = \tau \mu. \tag{B.13}$$

These choices satisfy every normalized constraint:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q, \qquad \tau + \bar{\mu}^\top z &= \tau(1 + \mu^\top z) = 1, \\ A_k(\Pi) - \theta_k \mu \leq 0 \implies (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ &= 0, \\ H(\bar{\Pi}) = \tau H(\Pi) &= \tau B > 0. \end{aligned}$$

Thus an attained optimal normalized solution cannot have $\tau = 0$.

Now take any normalized feasible point with $\tau > 0$ and define $\Pi = \bar{\Pi}/\tau$, $\mu = \bar{\mu}/\tau$. The row and movement constraints transform as follows:

$$\begin{aligned} \bar{\Pi}\mathbf{1} = \tau q &\iff \Pi\mathbf{1} = q, \\ A_k(\bar{\Pi}) &= \tau A_k(\Pi), \\ (A_k(\bar{\Pi}) - \theta_k \bar{\mu})_+ &= \tau (A_k(\Pi) - \theta_k \mu)_+, \\ \sum_j (A_{kj}(\bar{\Pi}) - \theta_k \bar{\mu}_j)_+ \leq \tau \theta_k &\iff \sum_j (A_{kj}(\Pi) - \theta_k \mu_j)_+ \leq \theta_k. \end{aligned} \tag{B.14}$$

The third line uses $(\tau t)_+ = \tau t_+$ for $\tau > 0$. Normalization then gives the denominator and the objective:

$$\begin{aligned} \tau + \bar{\mu}^\top z = 1 \iff \tau(1 + \mu^\top z) = 1 \iff \tau = (1 + \mu^\top z)^{-1}, \\ H(\bar{\Pi}) = \tau H(\Pi) = \frac{H(\Pi)}{1 + \mu^\top z}. \end{aligned} \tag{B.15}$$

The denominator is positive since $\mu, z \geq 0$. Conversely, start with any pair satisfying (6), choose τ as in (B.15), and multiply both objects by τ. Equation (B.14) verifies the row and movement conditions, and $\tau(1 + \mu^\top z) = 1$ verifies normalization. The objectives agree in both directions. This proves (7) and its attained maximum.

B.5 Equality on binary selections

For an original finite feasible point, put $\eta = J$ and $w_k = \lambda_k z$. Then

$$\begin{aligned} 0 \leq w_k \leq \lambda_k \mathbf{1}, \\ \theta_0 w_{0j} + \theta_1 w_{1j} = z_j \theta^\top \lambda \leq z_j J = z_j \eta, \\ a_i + \mathbf{d}_{i\ell}^\top w_{c_i} = a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z. \end{aligned} \tag{B.16}$$

The middle inequality is (3). All constraints of (4) therefore hold, giving $v(z) \leq V(z)$.

Conversely, for a risk-model feasible point, $\theta_k > 0$ and nonnegative prices give

$$\begin{aligned} z_j = 0 &\implies \theta_0 w_{0j} + \theta_1 w_{1j} \leq 0 \implies w_{0j} = w_{1j} = 0, \\ z_j = 1 &\implies w_{kj} \leq \lambda_k, \\ &\implies \quad w_k \leq \lambda_k z. \end{aligned} \tag{B.17}$$

Since $\mathbf{d}_{i\ell} \geq 0$ coordinatewise,

$$\Phi(y_i u_\ell) \leq a_i + \mathbf{d}_{i\ell}^\top w_{c_i} \leq a_i + \lambda_{c_i} \mathbf{d}_{i\ell}^\top z.$$

Thus (a, u, λ) is feasible in (2), with $V(z) \leq J \leq \eta$. Taking the minimum over risk-model feasible points gives $V(z) \leq v(z)$. Combining both directions yields $v(z) = V(z)$.

B.6 Convexity and completion of Theorem 2

Fix one feasible pair (Π, μ) in (6). Its remaining loss $h = H(\Pi) \geq 0$ is constant when z changes. For $z \in [0,1]^D$, define the scalar correction $r(z) = 1 + \mu^\top z > 0$ and its lower function $f(z) = h/r(z)$. For two coordinates $j, m \in \{1, \ldots, D\}$, the chain rule gives

$$\begin{aligned} \frac{\partial r}{\partial z_j} = \mu_j, \qquad \frac{\partial f}{\partial z_j} = h[-r(z)^{-2}]\mu_j = -\frac{h\mu_j}{r(z)^2}, \\ \frac{\partial^2 f}{\partial z_m \partial z_j} = -h\mu_j[-2r(z)^{-3}]\mu_m = \frac{2h\mu_j\mu_m}{r(z)^3}, \\ \nabla^2 f(z) = \frac{2h\mu\mu^\top}{r(z)^3}. \end{aligned} \tag{B.18}$$

For any direction $\xi \in \mathbb{R}^D$,

$$\xi^\top \nabla^2 f(z) \xi = \frac{2h}{r(z)^3}(\mu^\top \xi)^2 \geq 0.$$

Thus f is convex. Taking the maximum over the fixed feasible pairs in (7) proves that v is convex.

Together with the attained ratio representation and binary equality above, this completes the proof of Theorem 2.

C Proof of Theorem 3: reuse and quantitative convergence

The proof uses the same affine lower model as the global master. Extra tangents formed from retained fractions can strengthen it at any time. Whenever the guard accuracy is used, the candidate comes from the current model; earlier global lower bounds remain valid between guard solves.

C.1 Valid history bounds and bounded corrections

For each full binary evaluation s, retain a genuine feasible test at z^s with original finite score and prices (u^s, λ^s), objective $b_s = J(a^s, \lambda^s) \le B$, and some retained valid affine cut L_s satisfying $b_s - L_s(z^s) \le \delta$. The test or cut may come from earlier history: only their feasibility, anchor gap and bounded nonpositive cut gradient are required. Finite feasibility gives

$$\Phi(y_i u_\ell^s) \le a_i^s + \lambda_{c_i}^s \mathbf{d}_{i\ell}^\top z^s.$$

At a different binary z, define the repriced allowances by

$$a_i^s(z) = \max_\ell \{\Phi(y_i u_\ell^s) - \lambda_{c_i}^s \mathbf{d}_{i\ell}^\top z\}.$$

The definition implies $\Phi(y_i u_\ell^s) \le a_i^s(z) + \lambda_{c_i}^s \mathbf{d}_{i\ell}^\top z$ for every i, ℓ. Thus the finite-formulation theorem constructs an original feasible test, using this z, with risk at most $U^s(z) = q^\top a^s(z) + \theta^\top \lambda^s$. At the original selection, $a_i^s(z^s) \le a_i^s$, so $U^s(z^s) \le b_s$. A constant test supplies the bound B for every z. Taking the smallest stored bound gives $V(z) \le U_t(z)$, together with a corresponding test.

Define

$$D_{kj} = \sum_{i \in I_k} q_i \max_\ell d_{i\ell,j}, \qquad M_j = \max_k D_{kj}/\theta_k.$$

For a row-feasible matrix Π, its movement satisfies

$$0 \le A_{kj}(\Pi) = \sum_{i \in I_k, \ell} \Pi_{i\ell} d_{i\ell,j} \le \sum_{i \in I_k} \Big(\sum_\ell \Pi_{i\ell} \Big) \max_\ell d_{i\ell,j} = D_{kj}.$$

If a certificate has $\mu_j > M_j$, replace that component by M_j. For both classes, $A_{kj} - \theta_k M_j \le 0$, so the positive part in the certificate constraint remains zero in that coordinate. All other coordinates are unchanged. The new certificate remains feasible and its fraction does not decrease on $[0,1]^D$. Hence every retained certificate may be chosen with $0 \le \mu_j \le M_j$.

Choosing score zero in the definition of Ψ gives

$$0 \le h = H(\Pi) \le \Phi(0) \sum_\ell (p_{0\ell} + p_{1\ell}) = 2\Phi(0) = B.$$

Consequently every retained fraction has

$$\frac{\partial f(z)}{\partial z_j} = -\frac{h\mu_j}{(1 + \mu^\top z)^2}, \qquad -BM_j \le \frac{\partial f(z)}{\partial z_j} \le 0.$$

Its tangent is globally valid by Theorem 2, has a nonpositive gradient with the same componentwise bounds, and is tight to its own fraction at the tangent center. This applies equally to an anchor tangent and to every inexpensive recentered tangent. Therefore $m_t(z) \le V(z)$ on binary selections.

Minimizing m_t gives a lower bound on OPT. A globally valid solver lower bound is no larger than this minimum, and taking the maximum of all such bounds remains valid. Every incumbent comes from an original feasible test. These observations prove (10), including when a solve is interrupted. An incumbent master objective alone is not a global lower bound.

C.2 The one-sided history-width estimate

The own-source finite constraints give $a_i^s \geq 0$. Since the stored objective b_s is no larger than B,

$$\theta^\top \lambda^s \leq b_s \leq B.$$

For two selections z, z', each candidate inside the maximum defining a_i^s changes by $-\lambda_{c_i}^s \mathbf{d}_{i\ell}^\top (z - z')$. Only removed coordinates can increase that maximum, so

$$a_i^s(z) - a_i^s(z') \leq \lambda_{c_i}^s \sum_j \max_\ell d_{i\ell,j}\,(z_j' - z_j)_+.$$

After weighting by q_i and summing,

$$\begin{aligned} U^s(z) - U^s(z') &\leq \sum_j \sum_k \lambda_k^s D_{kj}(z_j' - z_j)_+ \\ &\leq (\theta^\top \lambda^s) \sum_j M_j(z_j' - z_j)_+ \leq B \sum_j M_j(z_j' - z_j)_+. \end{aligned}$$

This also proves coordinatewise nonincrease of the transferred upper bound.

Let L_s be the affine cut retained at evaluation s. It need not be the new certificate's tangent at z^s. Its bounded nonpositive gradient gives

$$L_s(z^s) - L_s(z) \leq B \sum_j M_j(z_j - z_j^s)_+.$$

The required accuracy of the retained pair is $b_s - L_s(z^s) \leq \delta$; since $U^s(z^s) \leq b_s$, $U^s(z^s) - L_s(z^s) \leq \delta$. For every stored evaluation s,

$$\begin{aligned} W_t(z) &= U_t(z) - m_t(z) \leq U^s(z) - L_s(z) \\ &= [U^s(z) - U^s(z^s)] + [U^s(z^s) - L_s(z^s)] + [L_s(z^s) - L_s(z)] \\ &\leq \delta + B \sum_j M_j\big[(z_j^s - z_j)_+ + (z_j - z_j^s)_+\big] \\ &= \delta + B\rho(z, z^s). \end{aligned}$$

Taking the minimum over the stored evaluations proves the history-width estimate in the theorem. The two one-sided changes occupy opposite directions; this is why their sum has coefficient B.

C.3 The global guard and admissible acquisition

For the current lower model, suppose the master returns a binary candidate z_M and certified bound ℓ with $m_t(z_M) - \ell \leq \xi$. The running bound satisfies $\mathrm{LB} \geq \ell$, hence $m_t(z_M) \leq \mathrm{LB} + \xi$. Repricing before the next full evaluation ensures $\mathrm{UB} \leq U_t(z_M)$. Thus

$$G = \mathrm{UB} - \mathrm{LB} \leq U_t(z_M) - m_t(z_M) + \xi = W_t(z_M) + \xi \leq \delta + \xi + B \min_s \rho(z_M, z^s).$$

This proves the guard-gap estimate. If $G > \varepsilon$ and $\xi \leq (1 - \alpha)\varepsilon$, then

$$W_t(z_M) \geq G - \xi \geq G - (1 - \alpha)\varepsilon > \alpha G,$$

and

$$m_t(z_M) \leq \mathrm{LB} + \xi < \mathrm{LB} + G = \mathrm{UB}.$$

The guard therefore meets both candidate conditions. Between scheduled guard solves, the previous running lower bound remains valid, and any eligible proposal may be evaluated. If no proposal is eligible, the guard is immediately refreshed. The acquisition rule may replace a guard with any eligible proposal, but a lack of such proposals cannot stall the algorithm. No probabilistic assumption about the proposal mechanism is used. If required solver accuracy cannot be obtained, the algorithm reports that exit with its valid bounds instead.

C.4 Packing and the number of full evaluations

Let $z^1, \dots, z^N$ be the completed accuracy-controlled binary evaluations; any numerical refinements needed at one selected point belong to that same evaluation. Fractional root calls are not included. Let G_N be the running gap after the Nth evaluation. The running upper bound can only decrease and the running lower bound can only increase, so the published gaps are nonincreasing.

For $i < j$, evaluation i is already in the history when z^j is chosen. Write G_j^- for the gap immediately before that call. The candidate condition and the history-width estimate imply

$$\alpha G_j^- < W_{j-1}(z^j) \le \delta + B\rho(z^j, z^i).$$

Since $G_j^- \ge G_N$,

$$\rho(z^j, z^i) > \frac{\alpha G_N - \delta}{B}.$$

When $G_N > \delta/\alpha$, this is a positive separation scale. By the definition of the packing number,

$$N \le \mathcal{P}\left(\mathcal{Z}_d, \rho, \frac{\alpha G_N - \delta}{B}\right).$$

Every call that begins while the gap exceeds ε instead satisfies, for each previous evaluation,

$$\rho(z^j, z^i) > r_\varepsilon := \frac{\alpha\varepsilon - \delta}{B} > 0.$$

There can therefore be no more than $\mathcal{P}(\mathcal{Z}_d, \rho, r_\varepsilon)$ such calls. If the algorithm were still continuing afterwards, the refreshed guard would be eligible and would require one more r_ε-separated point, a contradiction. Under the stated solve assumptions the algorithm must reach $G \le \varepsilon$. For the retained feasible test with actual risk R,

$$R \le \mathrm{UB} \le \mathrm{LB} + \varepsilon \le \mathrm{OPT} + \varepsilon.$$

This proves quantitative termination and the returned-test guarantee.

C.5 Interpreting the data-dependent complexity

Suppose an independently justified estimate $\mathcal{P}(\mathcal{Z}_d, \rho, r) \le Cr^{-p}$, with $C, p > 0$, holds at the separation scales under consideration. For $G_N > \delta/\alpha$, the packing result gives

$$N \le C\left(\frac{B}{\alpha G_N - \delta}\right)^p, \qquad G_N \le \frac{\delta + B(C/N)^{1/p}}{\alpha}.$$

When $G_N \le \delta/\alpha$, the same displayed upper bound is immediate. This rate is conditional on the stated packing estimate; it is not a universal smoothness or polynomial-time assertion.

A directly verifiable estimate is available from the feature weights. If $J \subseteq \{1, \dots, D\}$ satisfies $\sum_{j \notin J} M_j \le r$, two binary selections that agree on J obey

$$\rho(z, z') = \sum_{j \notin J} M_j |z_j - z'_j| \le \sum_{j \notin J} M_j \le r.$$

An r-separated family thus has at most one selection for each pattern on J. A selection in $\mathcal{Z}_d$ uses at most d coordinates there, so

$$\mathcal{P}(\mathcal{Z}_d, \rho, r) \le \sum_{i=0}^{\min(d,|J|)} \binom{|J|}{i} \le 2^{|J|}.$$

Choose $r = r_\varepsilon$ and let J contain the coordinates with the s_ε largest weights. By definition its omitted weight sum is at most r_ε. Combining the pattern bound with the packing termination result and the trivial bound by $|\mathcal{Z}_d|$ gives

$$N_\varepsilon \le \min\left\{ |\mathcal{Z}_d|, \sum_{i=0}^{\min(d,s_\varepsilon)} \binom{s_\varepsilon}{i} \right\} \le 2^{s_\varepsilon}.$$

The quantity s_ε is determined by the sorted data movement weights and the requested accuracy, without an unknown packing constant. It does not measure statistical importance. The bound can be much smaller than the number of all feasible subsets when the omitted movement weights have a small total. For arbitrarily fine precision with large weights, it may approach exhaustive combinatorial complexity. This limitation follows from the problem geometry and is compatible with all preceding guarantees.

D Loss formulas for Section 6

Write $F(s) = p_0\Phi(-s) + p_1\Phi(s)$, so $\Psi(p_0, p_1) = \inf_s F(s)$. First take $p_0, p_1 > 0$; the axes are treated below. When the minimum is attained, s^* denotes a minimizing scalar score.

For half hinge, $\Phi(s) = (1+s)_+/2$. If $s > 1$, replacing it by 1 keeps $\Phi(-s) = 0$ and reduces $\Phi(s)$. If $s < -1$, replacing it by -1 keeps $\Phi(s) = 0$ and reduces $\Phi(-s)$. It is therefore sufficient to minimize on $[-1, 1]$:

$$\begin{aligned} F(s) &= \frac{p_0(1-s) + p_1(1+s)}{2} = \frac{p_0+p_1}{2} + \frac{p_1-p_0}{2}s, \\ \Psi(p_0,p_1) &= \begin{cases} F(-1) = p_0, & p_1 \ge p_0, \\ F(1) = p_1, & p_1 < p_0, \end{cases} = \min(p_0, p_1). \end{aligned} \tag{D.1}$$

The choice of endpoint follows from the sign of the linear coefficient; when $p_0 = p_1$, every score in the interval is optimal.

For logistic, $\Phi(s) = \log(1 + e^s)$. Differentiate the two terms separately:

$$\begin{aligned} \frac{d}{ds}\left[p_0 \log(1+e^{-s})\right] &= -\frac{p_0 e^{-s}}{1+e^{-s}} = -\frac{p_0}{1+e^s}, \\ \frac{d}{ds}\left[p_1 \log(1+e^{s})\right] &= \frac{p_1 e^{s}}{1+e^{s}}, \\ F'(s) = \frac{-p_0 + p_1 e^s}{1+e^s} = 0 \quad &\Leftrightarrow \quad -p_0 + p_1 e^{s^*} = 0 \\ &\Leftrightarrow \quad e^{s^*} = p_0/p_1, \quad s^* = \log(p_0/p_1). \end{aligned}$$

The derivative changes from negative to positive at s^*, so this is the minimum. Substituting both logarithms gives

$$\begin{aligned} 1 + e^{-s^*} &= (p_0+p_1)/p_0, & 1 + e^{s^*} &= (p_0+p_1)/p_1, \\ \Psi(p_0,p_1) &= p_0 \log\frac{p_0+p_1}{p_0} + p_1 \log\frac{p_0+p_1}{p_1} \\ &= p_0[\log(p_0+p_1) - \log p_0] + p_1[\log(p_0+p_1) - \log p_1] \\ &= (p_0+p_1)\log(p_0+p_1) - p_0 \log p_0 - p_1 \log p_1. \end{aligned} \tag{D.2}$$

For exponential, $\Phi(s) = e^s$. A square proves the lower bound and identifies equality:

$$\begin{aligned} F(s) - 2\sqrt{p_0 p_1} &= \left(\sqrt{p_0}e^{-s/2} - \sqrt{p_1}e^{s/2}\right)^2 \geq 0, \\ \sqrt{p_0}e^{-s^*/2} = \sqrt{p_1}e^{s^*/2} \quad &\Leftrightarrow \quad e^{s^*} = \sqrt{p_0/p_1}, \quad s^* = \tfrac{1}{2}\log(p_0/p_1), \\ \Psi(p_0, p_1) &= F(s^*) = 2\sqrt{p_0 p_1}. \end{aligned} \tag{D.3}$$

For squared hinge, $\Phi(s) = (1+s)_+^2$. The same clipping argument restricts the score to $[-1, 1]$. Expanding and completing the square gives

$$\begin{aligned} F(s) &= p_0(1-s)^2 + p_1(1+s)^2 \\ &= (p_0 + p_1)s^2 + 2(p_1 - p_0)s + (p_0 + p_1) \\ &= (p_0 + p_1)\left(s - \frac{p_0 - p_1}{p_0 + p_1}\right)^2 + (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1}, \\ (p_0 + p_1) - \frac{(p_0 - p_1)^2}{p_0 + p_1} &= \frac{(p_0 + p_1)^2 - (p_0 - p_1)^2}{p_0 + p_1} = \frac{4p_0 p_1}{p_0 + p_1}, \\ s^* = \frac{p_0 - p_1}{p_0 + p_1} \in [-1, 1], \qquad &\Psi(p_0, p_1) = F(s^*) = \frac{4p_0 p_1}{p_0 + p_1}. \end{aligned} \tag{D.4}$$

The nonnegative square vanishes at this feasible score, which proves the minimum.

On the axes, nonnegativity and the indicated score limits give

$$\begin{aligned} 0 \leq \Psi(p_0, 0) \leq p_0\Phi(-s) \longrightarrow 0 \quad (s \to +\infty), \\ 0 \leq \Psi(0, p_1) \leq p_1\Phi(s) \longrightarrow 0 \quad (s \to -\infty). \end{aligned}$$

Consequently,

$$\Psi(p_0, 0) = \lim_{s \to +\infty} p_0\Phi(-s) = 0, \qquad \Psi(0, p_1) = \lim_{s \to -\infty} p_1\Phi(s) = 0, \qquad \Psi(0, 0) = 0.$$

For half hinge and squared hinge, the first zero is already attained at $s \geq 1$, and the second at $s \leq -1$. For logistic and exponential with a positive remaining mass, it is only an infimum as the score tends in the stated direction. The closed formulas use $0 \log 0 = 0$ and assign the squared-hinge value zero when both masses are zero.